

Jinhao Zhang

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EDUCATION:

Wuhan University (Wuhan, Hubei - China)

PhD. School of Water Resources and Hydropower Engineering

2025-Now

- Direct-entry PhD candidate at Wuhan University.
- Supervisor: Professor Jiyun Song.

Wuhan University (Wuhan, Hubei - China)

B.Eng. School of Water Resources and Hydropower Engineering

2021-2025

- Outstanding Undergraduate Graduation Thesis of Wuhan University.
- Supervisor: Professor Jiyun Song.

RESEARCH INTERESTS:

- Urban hydro-meteorological processes; Urban climate dynamics; Land-atmosphere interactions;
- Physics-driven urban microclimate simulation and prediction; Development and application of urban canopy model(UCM)
- Smart low-carbon city development: Multi-objective optimization for enhancing urban resilience

RESEARCH ARTICLES:

(*: Corresponding author)

- [1] Zhong Q, Song J*, Wang X, [Zhang J](#), Zhang Q, Li Y (2026), Modelling of lake-land breeze circulation in a diurnal cycle, *Building and Environment*, 293, 114319, DOI: 10.1016/j.buildenv.2026.114319.
- [2] [Zhang J](#), Song J*. Resilience of Urban Green Infrastructure to Compound Hot-Dry Extremes: Mechanisms and Water-Smart irrigation Strategies. In preparation.
- [3] Zhang Q, Song J*, Zhou Y, [Zhang J](#), Zhou Z. Physics-guided deep learning resolves five-metre urban thermal stress for climate-just delivery routing. Submitted to npj Urban Sustainability.

CONFERENCE:

- **Zhang J**, Song J* (2026), Modeling the Impact of Vegetation Dynamics on Urban Microclimate and Building Energy Demand. *European Geosciences Union 2026 (EGU)*, May 2026, Vienna, Austria. <https://doi.org/10.5194/egusphere-egu26-2289>. (Oral)
- **Zhang J**, Song J* (2025), Response Mechanisms of Urban Water-Heat Evolution Processes to Complex Underlying Surface Characteristics. *22nd China Water Forum*, September 2025, Xi'an, China. (Oral)
- **Zhang J**, Song J* (2025), Assessing the Cooling and Carbon Sequestration Potential of Urban Green Spaces via an Advanced Urban Canopy Model. *22nd Asia Oceania Geosciences Society (AOGS) Annual Meeting*, July 2025, Singapore. (Poster)
- **Zhang J**, Song J* (2025), Mechanisms of Urban Green Space Irrigation Impacts on the Water-Heat-Carbon Coupled System in the Built Environment. *12th International Conference on Urban Climate (ICUC12)*, July 2025, Rotterdam, Netherlands. <https://doi.org/10.5194/icuc12-156>. (Poster)
- **Zhang J**, Song J* (2025), Response Mechanisms of Urban Water-Heat Evolution Processes to Complex Underlying Surface Characteristics. *10th Youth Geoscience Forum*, May 2025, Hefei, China. (Oral)

EXCHANGE EXPERIENCE:

- 2025.11: The University of Hong Kong; Supporter: Jiyun Song, Jun Xia, Yuguo Li;
- 2025.08: National University of Singapore; Supporter: Jiyun Song;
- 2025.02: The University of Hong Kong; Supporter: Jiyun Song, Yuguo Li, Ji Chen;

RESEARCH EXPERIENCE:

- **PhD Research:**

2025-Now

Wuhan University, School of Water Resources and Hydropower Engineering
Dynamic Eco-Hydrological Responses of Urban Green Infrastructure to Seasonal Changes: Mechanisms and Water-Smart Irrigation Strategies:

- Using ensemble physical mechanisms and machine learning methods, this study assesses the resilience of urban green infrastructure
- Overcome the limitation of static vegetation parameterization in previous research by fully accounting for dynamic vegetation parameters.

AWARDS:

1. Outstanding Undergraduate Graduation Thesis of Wuhan University:
Zhang J, Song J*, *Response Mechanisms of Urban Hydrothermal Evolution to Complex Underlying Surface Characteristics*. (2025)
2. 2025 11th China Graduate Student Smart City Technology & Creative Design Competition, Second Prize of Wuhan University:
Zhang Q, **Zhang J**, Song J*, *River City Rider Guardian: A Dynamic Emergency System for Rider Heat Risk Based on Multi-source Data Fusion*. (2025)
3. Outstanding Youth Volunteers of Wuhan University. (2024)
4. Advanced Individual in Theoretical Learning of Wuhan University. (2024)
5. Outstanding Communist Youth League Cadre of Wuhan University. (2024)
6. Social Activity Activist of Wuhan University. (2023)
7. Wuhan University Excellent Pacemaker. Twice (2021,2024)
8. Advanced Individual of Golden Autumn in College of Water Resources and Hydropower of Wuhan University. (2021)
9. Tian Jiabing Scholarship. (2021)

UNDERGRADUATE EXPERIENCE:

- **Remote sensing observation and simulation of flooding in Dongdian flood storage.**
Research Assistant Sept. 2023 - Nov. 2023
Using ArcGIS and Google Earth, I manually labeled elevation and flood level at 3,000 points in Dongdian flood storage area to support satellite-based flood simulation.
- **Numerical modeling for water sciences (basin runoff forecasting)**
Participant Apr. 2024 - Jun. 2024
Using GIS data (DEM, watersheds, rivers, stations) and hydrometeorological data (water, rainfall, ET), we applied the Xinan River model to produce accurate hydrological forecasts.
- **Bishui Danxin, JiangZe interrogation practice activities**
Organizer Jun. 2022 – Sept. 2022
This practical activity was awarded the second prize of the College of Water Resources and Hydropower of Wuhan University.

SKILLS:

- **Urban microclimate simulation and prediction:** UCM
- **Programing Language:** MATLAB, Python, GAMS, SQL, C language, R language
- **Geospatial & Microclimate Analysis:** ArcGIS, CAD, Revit, Origin;
- **Language:** English(CET-6: 527); Mandarin(Level 1 Grade B)
- **Others:** Guitar, Hulusi, Taekwondo.